

Such administrators were placed in a team and their main task was to manage test and production environments, based on the rules of a particular team, with the resources allocated to the latter.

A DevOps engineer is, therefore, a specialist who implements the DevOps methodology in projects. He must have knowledge of different fields — he is a developer, a manager, a tester and a technical support specialist.

The DevOps engineer's responsibilities include:

- Deploying the release delivered by the developers in production
- Integration and deepening of development processes in the delivery
- Standardisation of the development environment
- Configuring the infrastructure for the features of the software being developed
- Preparing a productive environment for frequent changes
- Detecting and fixing problems
- Process automation

DevOps as a career path

Initially, DevOps had nothing to do with a specific position in the organisation. Many still claim that DevOps is a culture, not a profession, according to which communication between developers and system administrators should be established as closely as possible.

The developer should have an understanding of the infrastructure and be able to understand why a new feature running on a laptop suddenly brought half the data centre to a standstill. Such knowledge allows organisations to avoid conflicts — a programmer who is aware of how servers work will never dump all the responsibility on the sysadmin.

But what do recruiters and companies expect from a DevOps engineer? Often, they are expected to

have a mix of skills such as system administration, programming, using cloud technologies and automating large infrastructure.

This means that you need to not only be a good programmer, but also perfectly understand networks, operating systems, virtualisation, and how to ensure security and fault tolerance. You must also know several dozen different technologies, ranging from basic and time-tested things like iptables and SELinux to more recent and fashionable technologies like Chef, Puppet or even Ansible.

How to become a DevOps engineer

In general, a DevOps engineer is more about experience than knowledge of specific software. DevOps people are constantly learning, studying and testing new projects and technologies. They must constantly ask themselves: will this technology improve our project? What is better to choose as a language: Ruby, Python, Go, or write in pure pluses? And how will we deliver the changes to production, so as not to break the working systems?

The main thing to understand is that the DevOps specialist has a really good outlook. To expand it, you need to constantly engage in self-study.

Technical skills that are needed for DevOps

Senior systems administrator: The idea is that you should have a good understanding of the environment in which your applications will run. How they start (*init*, *systemd*), what to do if you see too many open files, use or not to use swap — these are issues you should be able to address.

Networking — CCNA: You can't write online services without understanding how the network works. No one says that you need to memorise the seven layers of the OSI model, but you will need to know exactly how IP, TCP/UDP and, of course, the application level protocol (for example, HTTP, HTTP/2) work. This will save you a lot of hassle when looking for the reasons for the *Connection Refused* error.

Junior developer: You have to understand how code is written, what OOP is, what threads are and a bunch of other things. In general, the more knowledge you have about all of these, the easier it will be for you to build and roll out the app.

Many people may disagree, arguing that the code should be written by the developer. But, if you don't understand how a software product is created, how will you automate its assembly, testing and deployment? Will you be able to notice a bottleneck in an architectural solution before it gets to production?

The year 2020 clearly showed that the transition to online and digital transformation for many companies is not only inevitable, but vital. Those who can switch to DevOps have a better chance of survival. Surely, it is not so easy to become a DevOps engineer from scratch. You will always have to study, read, and try. But after the 10th iteration, you will get a hold of the concept. The new and unknown is always cool. No matter what anyone says, go for it! **END** 🐧

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